

laptops need a portable power source to work unplugged. Packing more power in a small area is a key requirement in every consumer product today. As laptops, tablets and cellphones shrink in size and weight, without compromising on the quality of audio, graphics, etc, these need batteries which are thinner and lighter with higher power densities. Li-ion battery packs are much lighter and smaller than other battery chemistries, making them highly suitable for such applications.

Indian scenario. Despite being the second largest user of mobile phones in the world (which makes India a big consumer of Li-ion batteries too), India meets all of its Li-ion battery requirement through imports.

"There is no manufacturing facility in India till date to serve Li-ion cell in the commercial market. There are a few assemblers of Li-ion battery packs who import basic Li-ion cells from China, Korea and Japan. Li-ion battery pack assembly is done in India for the following products: mobile phones and tablets; handheld terminals like ticketing machines with printers, punching machines, credit card swipe machines, energy meter reading instruments and point-of-sale (POS) terminals; solar lanterns and other solar lighting applications; and vehicle tracking systems and weighing machines," informs S.S. Kandhari, managing director, Kandhari Photo Electronics.

Li-ion battery production is a very difficult and complex process, which needs a very clean environment as any kind of contamination may affect the battery's performance. Low-quality or duplicate Li-ion batteries being imported greatly harm the electronics. So there is need to promote use of standard and original batteries.

Kandhari informs, "Basically, there are two segments in assembly of Li-ion battery packs and two different categories of manufacturers. In consumer sector, there are a large number of mobile phone and tablet battery assemblers but only a few of them are doing quality business, i.e., using quality/branded cells and quality protection circuits for Li-ion battery pack assembly. Others

are doing cheap business and compromising on the quality of batteries by using B-grade unbranded cells and circuits.

"In industrial sector, there are very few assemblers who have deep knowledge about the battery and its controlling PCB nature, and better assembly and testing equipment in nature, it is necessary to use cells of a reputed brand and proper protection circuits."

Home-grown mobile phone manufacturer Maxx Mobile is one company which is going to produce its own batteries. The company had invested around ₹ 4 billion in setting up two units at Haridwar to produce mobile phone batteries and feature phones. But operations were stopped approximately two years ago, when a handset import duty of 1.34 per cent made domestic manufacturing unviable. However, now that the scene has changed, with the government's decision to levy seven per cent customs duty on handsets priced above ₹ 2000, Maxx is returning to produce its own products.

Opportunities and threats. Big opportunities exist as customers need to change over from sealed lead-acid batteries to Li-ion batteries due to environment issues.

"As green energy is now preferred, there will be no alternative for sealed lead-acid battery users but to shift to Li-ion in small (like torches and solar lanterns) as well as large applications. Also, nowadays all high-standard and basic consumer gadgets like tablets, smartphones and laptops run on Li-ion or Li-poly batteries due to their light weight and compact size," shares Kandhari.

However, to capitalise on this opportunity, it is important to remove some roadblocks. "Currently, government policies, taxes and duty structure do not favour manufacturing in India. Policies are only in books and there is no implementation. Surge in dollar price is a major cause of concern to assemblers, as for



3.7V, 850mAh Li-ion battery

Li-ion cells and protection PCBs they totally depend on imports. Also, small assemblers who don't have any battery knowledge and testing facility to check the battery are ruining the market by duplication and creating cheap competition in the market," adds Kandhari.

Future outlook. Kandhari says, "The government has started encouraging green environment policies as it is now the need of the country. If these policies get implemented, people will need to switch from sealed lead-acid batteries to Li-ion batteries; currently sealed lead-acid batteries account for 90 per cent of the market.

"Being light-weight, small-size and high-capacity, Li-poly battery is widely used in electronic gadgets like iPods, tablets and mobile phones. In fact, in the future, we can expect Li-poly batteries to grow faster than Li-ion batteries.

"The scope of a new battery technology called LiFePO₄ (lithium-iron phosphate) is also very big because of its following features:

1. Longer life of up to 2000 cycles (as against sealed lead-acid batteries' only 300 cycles and Ni-MH batteries' 500 cycles)
2. Higher capacity and voltage (one can install many single cells in series and in parallel, and the battery can fully discharge)
3. More powerful (high discharge current, 2C continuous, 5C pulse for normal cells and 25C for special design)
4. Safer (the battery never explodes and is stable in high temperatures, so no temperature controller is needed)
5. Lighter and smaller
6. Most environment-friendly among lithium batteries (no lead or cadmium used)
7. Rechargeable by solar power also"

However, Li-ion and Li-poly battery manufacturing can be started in India only if the government provides support for basic raw material and subsidies required for manufacturing of Li-ion and Li-poly batteries. ●

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